

Technical Note

Signal & Noise Model for PlatoSim

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1 Abstract

In this document we provide an overview on the algorithms that are currently used to simulate the signal and noise contributions in PlatoSim.

2 Documents and Acronyms

2.1 Applicable Documents

AD-1: PLATO-DLR-PL-RP-001, *Instrument Noise Budget*

AD-2: PLATO-KUL-PL-TN-0001, *PlatoSim Reference Frames*

AD-3: PLATO-KUL-PL-TN-0006, *Configuration Parameters for PlatoSim*

AD-4: PLATO-DLR-PL-RP-0004, *Simulations and performance estimations*

2.2 Reference Documents

RD-1: *An analytical model of radiation-induced Charge Transfer Inefficiency for CCD detectors*, Short, A. et al., MNRAS 420, 3078-3085, 2013

RD-2: *Evidence for self-interaction of charge distribution in charge-coupled devices*, Guyonnet, A. et al., A & A, 575, A41, 2015

2.3 Acronyms

ADC	Analogue-to-Digital Converter
ADU	Analogue-to-Digital Unit
BOL	Beginning Of Life
CCD	Charge-Couple Device
CVF	Charge Voltage (conversion) Factor
CTI	Charge Transfer Inefficiency
DSNU	Dark Signal Non-Uniformity
EOL	End Of Life
FEE	Front-End Electronics
FWHM	Full-Width at Half-Maximum
PRNU	Pixel-Response Non-Uniformity
PSF	Point Spread Function
QE	Quantum Efficiency

3 Introduction

This document aims at providing an overview on the exact algorithms and formulae that are being used to simulate the signal and noise contributions in PlatoSim. For a detailed description of how to use PlatoSim, we refer to the PlatoSim GitHub documentation pages. For further reading on the values of the configuration parameters for PlatoSim we refer to [AD-3].

4 Noiseless Signal

To determine the signal in any given pixel of the CCD, we chop up an exposure into smaller time steps and accumulate photons / electrons / ADUs over these small periods of time. The smaller time steps, referred to as the ‘heartbeat’ of the instrument, are necessary to correctly account for the effect of the spacecraft jitter and thermo-elastic instabilities of the optical bench (together referred to as ‘jitter’), which cause small shifts in star position over the detector.

In the absence of noise, the signal in pixel (i, j) ¹, expressed in ADU, can be written as:

$$S_{ij}(t) = \left[\left(\lfloor B \cdot t_{\text{exp}} \rfloor + \sum_{n=1}^{N_J} \sum_{k=1}^{K_*} [\Delta t_n \cdot F_k(t_n) \cdot f_{ij}(x_k(t_n), y_k(t_n))] \right) \cdot T_{\lambda, ij}(t) \cdot E_{ij} \cdot g_{ij}(t) + c(t) \right] \quad (1)$$

with

- t the start time of the exposure, [s];
- B the total sky background, [photons/pixel/s]
- t_{exp} the exposure time, [s];
- $\sum_{n=1}^{N_J}$ the sum over all jitter steps in the exposure;
- $\sum_{k=1}^{K_*}$ the sum over all stars in the FOV;
- Δt_n the duration of the n ’th jitter step, [s];
- F_k the flux received from the k ’th star in the Plato passband, [photons/s];
- t_n the time at the start of the n ’th jitter step, [s];
- $f_{ij}(x_k(t_n), y_k(t_n))$ the integral of the PSF at pixel (i, j) given that the k ’th star is at pixel (x_k, y_k) during the n ’th jitter step (see Sect. 7);
- $T_{\lambda, ij}$ the transmissivity of the optics at pixel (i, j)
- E_{ij} , the product of the efficiency factors at pixel (i, j) (see Sect. 8);
- $g(t_n)$ the temperature dependent gain (of the CCD and the FEE combined) [ADU/pixel] (see Sect. 9);
- and $c(t_n)$ the temperature dependent electronic offset; [ADU] (see Sect. 9).

In this formula, $\lfloor \dots \rfloor$ denotes ‘the integer part of’, as photon counts and ADUs are integers.

The flux received from each star can be computed with:

$$F_k(t) = F_0 \cdot W \cdot A \cdot 10^{-0.4m_k(t)}, \quad (2)$$

with

¹Subscript i refers to the column coordinate, subscript j to the row coordinate.

- F_0 the flux received from a 0'th magnitude star ($m=0$) in the Plato passband, [photons/s/cm²];
- W the throughput bandwidth (FWHM) of the Plato passband, [nm];
- A the effective light-collecting area of the telescope, [cm²];
- t_n the time at the start of the n 'th jitter step, [s];
- and $m_\lambda(t)$ the (possibly) time dependent magnitude of the star in the Plato passband.

PlatoSim simulates that both detector halves are read out separately and sent to their own readout register and front-end electronics (FEE). Rather than making simulations for the whole image area of 4510×4510 pixels, PlatoSim allows to simulate a smaller area of $N_{\text{rows},S} \times N_{\text{columns},S}$ pixels, which we refer to as the 'sub-field' (or 'pixel map').

In the following sections we explain how the different parts of Eq. (1) have been implemented in PlatoSim. Sects. 10 to 12 show how the contributions of various sources of noise come into play and are being taken into account in PlatoSim.

5 Time Dependent Effects

The following effects are time dependent in PlatoSim:

- The stellar magnitude $m(t) \equiv m_0 + \delta m(t)$ in the Plato passband can be time dependent. PlatoSim takes this into account via a user-provided time series $\delta m(t)$.
- The position of a star on the detector (x_k, y_k) slightly changes under influence of jitter. PlatoSim takes this into account via a user-provided time series of the yaw, pitch, and roll of the platform.
- Also thermo-elastic variations slowly change the position of the star on the detector. PlatoSim uses a user-provided time series of the yaw, pitch, and roll of the telescope to simulate a slow drift.
- Ageing effects cause the transmissivity of the optics T_λ to decrease over the course of the mission. PlatoSim uses a linear model with user-provided BOL and EOL values.
- Both the gain g and the electronic offset c are temperature dependent. PlatoSim takes this into account via a user-provided time series of the temperature variation.
- The focal length can slowly vary in time, causing the position of the stars on the CCD to change slowly. PlatoSim takes this into account using a user-provided time series of the focal-length values.
- The optical distortion can slowly vary in time, causing the position of the stars on the CCD to change slowly. PlatoSim takes this into account using a user-provided time series of the optical distortion coefficients.

6 Stellar Pixel Coordinates

6.1 Stellar Aberration

Before converting the sky coordinates to the position on the detector, stellar or kinematic aberration must be taken into account.

The input star catalogue provides star positions in equatorial coordinates (α, δ) . Kinematic aberration, however, is calculated for ecliptic coordinates (λ, β) . The first step is therefore converting from equatorial to ecliptic coordinates.

We then calculate the relativistic aberration angle between the velocity direction of the spacecraft and the star direction using the velocity-addition formula:

$$\tan \theta = \frac{\sqrt{1 - \beta^2} \sin \theta'}{\cos \theta' + \beta}, \quad (3)$$

where β is the ratio of the velocity of the spacecraft and the speed of light, and θ' is the angle between the velocity direction and the star direction.

The calculation is an approximation based on a circular earth orbit around the Sun at a constant speed of 30 km/s and not taking into account the orbit of the satellite around L2.

The differential kinematic aberration is the difference between the aberration correction with respect to the star direction and the aberration correction for the pointing direction. We rotate the aberrated star direction to compensate for aberrated pointing to get the differential aberrated star direction.

The ecliptic coordinates are then calculated for the new aberrated star direction and we convert back from ecliptic coordinates to equatorial coordinates.

6.2 From Sky to Pixel Coordinates

The undistorted position $(x_{\text{FP}}, y_{\text{FP}})$ of a star on the focal plane (in mm) is calculated by applying Eqs. (5), (6), and (8) from [AD-2] to the equatorial coordinates from the input catalogue, after having applied stellar aberration. How spacecraft jitter and the thermo-elastic instabilities of the optical bench (so-called drift) are taken into account, is explained in Sects. 8 and 9 in [AD-2].

The distorted focal-plane coordinates $(x'_{\text{FP}}, y'_{\text{FP}})$ are calculated (in mm) using:

$$\begin{cases} x'_{\text{FP}} = x_{\text{FP}} + \cos \alpha \cdot \Delta r_{\text{FP}} \\ y'_{\text{FP}} = y_{\text{FP}} + \sin \alpha \cdot \Delta r_{\text{FP}} \end{cases} \quad (4)$$

where α is the position angle of the star

$$\alpha = \text{atan}(y_{\text{FP}}/x_{\text{FP}}). \quad (5)$$

and the distortion Δr_{FP} of the radial distance is computed by

$$\Delta r_{\text{FP}} = [k_1 \cdot r_{\text{FP}}^3 + k_3 \cdot r_{\text{FP}}^5 + k_5 \cdot r_{\text{FP}}^7] \cdot f, \quad (6)$$

with r_{FP} the normalised radial distance from the optical axis:

$$r_{\text{FP}} = \frac{\sqrt{x_{\text{FP}}^2 + y_{\text{FP}}^2}}{f} \quad (7)$$

where f is the focal length in mm.

Subsequently, these distorted focal-plane coordinates are converted to pixel coordinates on the detector using Eq. (14) in [AD-2].

7 The Point-Spread Function

There are two options to take the PSF into account:

1. Convolution of the sub-field with a pre-computed PSF.
 - Pro: the convolution is done at sub-pixel level (at arbitrary resolution), and the intra-pixel response non-uniformity (IRNU) can therefore be taken into account.
 - Con #1: The PSF is fixed and is therefore not dependent on the radial distance of the optical axis.
 - Con #2: A convolution always implies edge effects. The solution is to select a larger sub-image and cut out the non-affected part.

Table 1: Overview of which efficiency factor are a fixed number over the FOV and which ones depend on the position in the FOV.

Efficiency factor	Fixed number	Position dependent
Quantum efficiency	✓	✓ (angular radial distance from the optical axis)
Vignetting		
Polarisation	✓ (for now)	
Particulate contamination	✓	
Molecular contamination	✓	✓ (distance from readout register)
Charge-transfer inefficiency		
Angular dependency of the QE	✓ (for now)	

2. The use of an analytic PSF (given in [AD-4]).

- Pro: The dependence of the PSF on the radial distance from the optical axis is taken into account.
- Con: To be reasonably fast, the analytic model should be computed at pixel level, so that the IRNU cannot be taken into account.

8 Efficiency Factors

Various effects come into play, further decreasing the efficiency of the set-up. These are enclosed in the factor E_{ij} in Eq. (1). The number of photons or electrons is multiplied with each of these efficiency factors to account for these losses. Some of the effects are (currently considered) uniform over the whole FOV (hence we multiply the number of photons with a fixed number for those effects), whilst others are dependent on the angular radial distance θ of the pixel (*row, column*) (in the CCD reference from) to the optical axis or on the distance to the readout register, as shown in Table 1.

We refer to Sects. 6 and 7 of [AD-2] for the calculation of θ from the position in the CCD reference frame. Below we discuss the efficiency factors that are position dependent.

8.1 Vignetting

The brightness attenuation towards the edges of the FOV is contributed to by natural and mechanical vignetting, only the former of which is simulated by PlatoSim. The efficiency caused by natural vignetting for pixel (i, j) (in the CCD reference frame) can be approximated by

$$E_{\text{vig.}}(i, j) = \cos^2(\theta), \quad (8)$$

where θ is the angular radial distance to the optical axis for pixel (i, j) .

8.2 Charge-Transfer Inefficiency (CTI)

Simple Approach In the simplest approach we make a distinction between the CTI in the parallel direction (CTI_{parallel}) and in the serial direction (CTI_{serial}). The efficiency due to CTI for pixel (i, j) is then:

$$E_{\text{CTI}}(i, j) = (1 - CTI_{\text{parallel}})^{j+1} \cdot (1 - CTI_{\text{serial}})^{i+1} \quad (9)$$

for the left half of the CCD ($j < 2255$) and

$$E_{\text{CTI}}(i, j) = (1 - CTI_{\text{parallel}})^{j+1} \cdot (1 - CTI_{\text{serial}})^{4510-i+1} \quad (10)$$

for the right half of the CCD ($j \geq 2255$).

Short et al. 2013 An alternative implementation is based on [RD-1] (Sect. 3.2 - 3.3). We start by calculating the α as in Eq. (23) in [RD-1], for each trap species k :

$$\alpha(k) = \frac{t_{\text{transfer}} \cdot \sigma(k) \cdot v_t \cdot FWC^\beta}{2 \cdot V_g}, \quad (11)$$

where t_{transfer} is the charge transfer time², $\sigma(k)$ is the trap capture cross-section for trap species k , v_t is the electron thermal velocity as calculated in Eq. (9) in [RD-1]³, FWC is the full-well capacity of the detector, and V_g is the maximum geometrical volume that electrons can occupy within a pixel:

$$V_g = \frac{(18 \cdot 10^{-6})^2 \cdot 10^{-6}}{2} \text{m}^3. \quad (13)$$

We assume there are initially no occupied traps in any column i for any trap species k :

$$N_o(i; k) = 0. \quad (14)$$

For every column i and every trap species, we follow the following procedure for all rows j in the sub-field:

- Calculate γ as in Eq. (23) in [RD-1]:

$$\gamma(i; k) = \frac{S_{ij} \cdot n_t(k) \cdot V_g}{FWC^\beta} \cdot i, \quad (15)$$

where $n_t(k)$ is the trap density for trap species k .

- Calculate the number of captured electrons for that pixel with Eq. (22) in [RD-1]:

$$N_c(i, j; k) = \max \left\{ \frac{\gamma(i; k) \cdot S_{ij}^\beta - N_o(i; k)}{\gamma(i; k) \cdot S_{ij}^{\beta-1} + 1} \cdot \left(1 - \exp \left(\alpha \cdot S_{ij}^{1-\beta} \right) \right), 0 \right\}. \quad (16)$$

- Update the number of occupied traps:

$$N_o(i; k) \rightarrow N_o(i; k) + N_c(i, j; k). \quad (17)$$

- Calculate the number of released electrons N_r with Eq. (25) in [RD-1]:

$$N_r(i, j; k) = N_o(i; k) \cdot \left(1 - \exp \left(-\frac{t_{\text{transfer}}}{\tau_r} \right) \right), \quad (18)$$

where τ_t is the trap release time constant for that particular trap species.

- Update the number of occupied traps:

$$N_o(i; k) \rightarrow N_o(i; k) - N_r(i, j; k). \quad (19)$$

- Add the excess electrons to the current pixel:

$$S_{ij} \rightarrow S_{ij} + (N_r(i, j; k) - N_c(i, j; k)). \quad (20)$$

²The charge transfer time is calculated as

$$v_{\text{transfer}} = \frac{t_{\text{readout}}}{4510}, \quad (12)$$

which is the time needed to shift down the charges by one row.

³The temperature T to calculate the electrons' thermal velocity is configurable separately from the operating temperature of the detector. The free electron mass, used to calculate the effective electron mass m_e^* is $9.10938215 \cdot 10^{-31} \text{kg}$.

8.3 Angular Dependency of the QE

Currently only the mean angle dependency of the quantum efficiency is used instead of a true angle dependent factor (Anko Börner, priv. comm.).

9 Signal Conversion

The conversion from electrons per pixel to ADU is done using

$$S_{ij}[\text{ADU}] = S_{ij}[\text{electrons}] \cdot g_{\text{CCD}} \cdot g_{\text{FEE}} + c, \quad (21)$$

where g_{CCD} is the CCD gain (expressed in $\mu\text{V}/\text{electron}$), g_{FEE} is the FEE gain (expressed in $\text{ADU}/\mu\text{V}$), and c is the electronic offset. Different gain values may be used for both detector halves.

The gain values and the electronic offset are simulated as temperature dependent:

$$\begin{aligned} g_{\text{CCD}}(T_{\text{CCD}}) &= g_{\text{CCD}}(T_{\text{CCD,nominal}}) + (T_{\text{CCD}} - T_{\text{CCD,nominal}}) \cdot \Delta g_{\text{CCD}}, \\ g_{\text{FEE}}(T_{\text{FEE}}) &= g_{\text{FEE}}(T_{\text{FEE,nominal}}) + (T_{\text{FEE}} - T_{\text{FEE,nominal}}) \cdot \Delta g_{\text{FEE}}, \\ c(T_{\text{FEE}}) &= c(T_{\text{FEE,nominal}}) + (T_{\text{FEE}} - T_{\text{FEE,nominal}}) \cdot \Delta c_{\text{FEE}}, \end{aligned} \quad (22)$$

where T_{CCD} and T_{FEE} are the CCD resp. FEE operating temperatures. The slopes and the intercepts of the linear relations stated above are given by the user.

10 Astrophysical Noise

10.1 Cosmic Hits

The number of cosmits hitting an area of $n_{\text{rows}} \times n_{\text{columns}}$ pixels is computed as

$$N_c = \mathcal{P}(h_0) \cdot \Delta t \cdot n_{\text{rows}} \cdot n_{\text{columns}} \cdot s_p^2 \quad (23)$$

where $\mathcal{P}(h_0)$ is the Poisson distributed cosmic hit rate with (a user-given) mean h_0 (events / cm^2 / s) and s_p is the pixel size. For the image area, Δt is the exposure time plus the readout time, for the parallel over-scan only the readout time.

The entry points for the cosmic hits are sampled uniformly over the image area or the parallel over-scan. The length of the trail is drawn from a uniform distribution over user-given interval, and the same holds for the number of electrons that are dumped by the cosmic. The orientation angle is distributed uniformly over the interval $[0^\circ, 360^\circ]$. The excess electrons induced by a cosmic hit are distributed over the trail length, according to a decay function f :

$$f(i) = \frac{\exp\left(\frac{-d_i^2}{2 \cdot \sigma^2}\right)}{\sum_{i=0}^{199} \exp\left(\frac{-d_i^2}{2 \cdot \sigma^2}\right)}, \quad (24)$$

where d_i are 200 equally spaced points along the trail, and $\sigma = l/3$. The excess electrons that fall off the detector are lost. We only add cosmits to the image area and the parallel over-scan, and neglect them for the serial pre-scan (as these elements are short-lived).

Note that in case of partial readout, no parallel over-scan will be performed.

10.2 Photon Noise

The contribution of the photon noise that is added to a pixel's value (in the image area and the parallel over-scan) is drawn from a Poisson distribution with the pixel value as mean:

$$S_{ij}[\text{e-}/\text{pixel}] \xrightarrow{\text{photon noise}} \mathcal{P}(S_{ij}). \quad (25)$$

Note that in case of partial readout, no parallel over-scan will be performed.

11 CCD Related Noise

11.1 Dark Current

Dark-signal non-uniformity (DSNU) is a fixed-pattern deviation of a pixel's dark current from its nominal value and is characterised by a percentage of the nominal value. Dark signal is added to the image area and the parallel over-scan. For the former, the dark-signal is given by

$$S_{ij,\text{dark}} = \mathcal{N}\left(\mu = n_{DS} \cdot t, \sigma = n_{DS} \cdot t \cdot \frac{DSNU}{100}\right), \quad (26)$$

where n_{DS} is the dark-signal rate, and t is the time needed for the exposure and the readout. Note that for the fast camera the lower half of the CCD is read out (after frame transfer) whilst the next exposure is already taking place. Also during this part of the readout dark current is acquired.

If we also include the dark-signal shot noise, we have:

$$S_{ij} \xrightarrow{\text{dark signal (incl. DSNU and noise)}} S_{ij} + \mathcal{N}(\mu = S_{ij,\text{dark}}, \sigma = \sqrt{S_{ij,\text{dark}}}). \quad (27)$$

Note that in case of partial readout, no parallel over-scan will be performed.

11.2 Pixel-Responsivity Non-Uniformity (PRNU)

Pixel-response non-uniformity is a fixed-pattern deviation of a pixel's responsivity from its nominal value. In case of an analytical PSF, the PRNU is generated at pixel level; for pre-computed PSFs or PSFs that are generated on-the-fly, the PRNU is generated at sub-pixel level (see Sect. 7).

Assume a sub-field of $n_{\text{rows}} \times n_{\text{columns}}$ pixels or sub-pixels (depending on which PSF you are using). The PRNU map must have the same dimensions. It is generated by the following procedure:

- Generate an empty 2D array, the dimensions of which are double of those of the sub-field (i.e. $2 \cdot n_{\text{rows}} \times 2 \cdot n_{\text{columns}}$). This is needed because of the behaviour of the Fourier transforms.
- For all elements (i, j) in this array (counting starting at 0), generate white noise (as a sample from a normal distribution $\mathcal{N}(0, 1)$) and include a $1/f$ dependency by dividing by $i^2 + j^2 + 1$.
- Take the real part of the inverse Fourier transform.
- Cut out a part of the map by the origin, as big as the sub-field.
- Normalise the map to the $[0, 1]$ interval (by subtracting the minimum value in the array and dividing by the range).
- Multiply by the given peak-to-peak pixel noise and subtract by 1 minus the peak-to-peak pixel noise to map it onto the interval $[1 - \text{peak-to-peak amplitude}, 1]$.

After doing the calculating what is between square brackets in Eq. (1), we multiply the sub-field with the flatfield map we generated as described above:

$$S_{ij} \xrightarrow{\text{PRNU}} S_{ij} \cdot PRNU_{ij}. \quad (28)$$

11.3 Brighter-Fatter Effect (BFE)

The brighter-fatter effect is implemented according to the method proposed in [RD-2]. Using the sub-scripting nomenclature from [RD-2] (this section only), we consider any pixel $(0,0)$ with charge Q_{00} , which feels the influence of the charges Q_{ij} accumulated in the pixels (i,j) in a square window of a fixed sized $w = 2 + 1$ centred around it, shifting the boundaries with its neighbouring pixels $X \in \{(0,1), (0,-1), (1,0), (-1,0)\}$ with charge Q_X .

We can calculate the perturbed charge in pixel $(0,0)$ with Eq. (11) in [RD-2]:

$$Q_{00} \xrightarrow{BFE} Q_{00} + \frac{1}{4} \sum_X \sum_{i,j} a_{ij}^X \cdot Q_{ij} \cdot (Q_{00} + Q_X), \quad (29)$$

where the a_{ij}^X coefficients for the pixels (i,j) in a square window centred around $(0,0)$ ($i, j \in [-hw, hw]$) are

$$a_{ij}^X = p_0 \cdot Ei(p_1 \cdot r_{ij}) \cdot \cos \theta_{ij}^X. \quad (30)$$

The parameters r_{ij} and $\cos \theta_{ij}^X$ in this equation are the distance between the charge in pixel (i,j) and the boundary of pixel $(0,0)$ with each of the neighbours X , and the cosine of the angle between the source-boundary vector and the normal to the boundary resp., and can be calculated from the following set of equations:

$$\Delta i = i - \frac{X[0]}{2}, \quad (31)$$

$$\Delta j = j - \frac{X[1]}{2}, \quad (32)$$

$$r_{ij} = \sqrt{(\Delta i)^2 + (\Delta j)^2}, \quad (33)$$

$$\cos \theta_{ij}^X = \frac{(\Delta i, \Delta j) \bullet X}{r}, \quad (34)$$

where $\bullet$ denotes the dot product.

Term Ei in Eq. (30) is the exponential integral:

$$Ei(x) \equiv \int_{-x}^{\infty} \frac{e^{-t}}{t} dt; x > 0. \quad (35)$$

Enforcement of the sum rule for the a_{ij}^X coefficients (Eq. (7) from [RD-2]) is obtained by subtracting $\sum_X a_{ij}^X$ from the a_{ij}^X coefficients from Eq. (30).

11.4 CCD Readout Noise

The readout noise of the CCD and of the FEE are treated together and are added in quadrature. Consider $\mu_{\text{RON,CCD}}$ and $\mu_{\text{RON,FEE}}$ the mean readout noise for the CCD and FEE resp. The contribution of the readout noise is sampled from a normal distribution and added to the signal (in the image area, serial pre-scan, and parallel over-scan):

$$S_{ij} \xrightarrow{\text{readout noise}} S_{ij} + \mathcal{N}\left(0, \sqrt{\mu_{\text{RON,CCD}}^2 + \mu_{\text{RON,FEE}}^2}\right). \quad (36)$$

11.5 Open-Shutter Smearing

Due to the absence of a shutter, the detector still receives light during readout. Consequently, the flux of a pixel gets affected by the flux of the pixels in the same column. For the fast camera, the lower half of the CCD

is shielded off against incoming radiation and hence we therefore have a contribution during the frame transfer (and not during the readout of the lower half of the CCD, which is done whilst the next exposure is already taking place).

$$S_{ij}[\text{e}^-/\text{pixel}] \xrightarrow{\text{open shutter}} S_{ij} + \left(\sum_{j \in S} S_{ij} + B \cdot (4510 - \text{row}_0 - N_{\text{rows},S} + N_{PO}) \cdot \langle E \rangle \right) \cdot \frac{t_{\text{readout}}}{(4510 - \text{row}_0) \cdot t_{\text{exp}}}, \quad (37)$$

where the summation runs over all $N_{\text{rows},S}$ rows in the sub-field, N_{PO} is the number of rows in the parallel over-scan, row_0 is the first exposed row (0 for the normal camera, 2255 for the fast camera), and t_{readout} is the time needed for the whole readout process for the normal camera or the time needed for frame transfer for the normal camera. Term $\langle E \rangle$ is the product of the expected efficiencies due to quantum efficiency, vignetting, polarisation, and particulate and molecular contamination (see Sect. 8).

For the fast camera we account for the fact that the lower half of the detector is shielded off against incoming radiation.

11.6 Full-Well Saturation (Blooming)

If a pixel receives more electrons than the full-well saturation limit l_1 , the excess electrons flow evenly distributed in positive and negative charge-transfer direction but within the same column: half of the excess electrons are pushed away from the readout register, the other half towards it, but still within the same column as the saturated pixel. Electrons reaching the edge of the CCD will not be detected. Care must be taken not to saturate pixels by moving excess electrons to it.

The pseudocode of the implementation of blooming in PlatoSim can be found in Algorithm 1.

12 FEE Related Noise

12.1 FEE Readout Noise

See Sect. 11.4.

12.2 Digital Saturation

Digital saturation is implemented as topping off the signal (in the image area, serial over-scan, and parallel over-scan) at the digital saturation limit l_2 :

$$S_{ij} \xrightarrow{\text{digital saturation}} \min(S_{ij}, l_2). \quad (38)$$

13 Serial Pre-Scan and Parallel Over-Scan

At the end of the readout registers closest to the readout electronics there are N_{SP} (typically 25) extra elements, composing the serial pre-scan (or bias register map). After all rows of the image area have been read out (in time span t_{readout}), N_{PO} (typically 30) virtual rows are read out, hence composing the parallel over-scan (or smearing map). Table 2 shows which effects PlatoSim takes into account for each scan.

Algorithm 1: Pseudocode for the implementation of blooming. If a pixel exceeds the full-well saturation limit, the excess electrons will flow evenly up and down the column (i.e. half of the excess electrons flows towards the readout register, the other half away from it).

```

1 for  $j \leftarrow 0$  to 4509 do
2   for  $i \leftarrow 0$  to 4509 do
3     if  $S_{ij} > l_1$  then
4       // Full-well saturation limit exceeded
5        $jmod \leftarrow j$ ; // Row coordinate where excess e- are transferred from and to
6        $numExcessElectrons \leftarrow \frac{S_{ij} - l_1}{2}$ ;
7       /* When we move half of the excess electrons down, the pixel below may also
8          become saturated (or was already saturated). When processing this pixel
9          below, we need to avoid that it also sends its excess electrons up and
10         down, and therefore sends part of its excess electrons back up where they
11         actually came from. The variable transfer2Saturated is to keep track
12         whether the destination pixel is also saturated, so that it does not send
13         back electrons where they came from. */
14        $transfer2Saturated \leftarrow false$ ;
15       while ( $numExcessElectrons > 0$ ) and ( $jmod < 4510$ ) do
16         if not  $transfer2Saturated$  then
17            $S_{i,jmod} \leftarrow S_{i,jmod} - numExcessElectrons$ ;
18            $jmod \leftarrow jmod + 1$ ;
19           // Electrons reaching the edge of the CCD will not be detected
20           if  $jmod < 4510$  then
21             if  $S_{i,jmod} \geq l_1$  then
22                $transfer2Saturated \leftarrow true$ ;
23             else
24                $transfer2Saturated \leftarrow false$ ;
25                $S_{i,jmod} \leftarrow S_{i,jmod} + numExcessElectrons$ ;
26               /* Make sure the pixel you move the excess electrons to does not get
27                  saturated too */
28               if  $S_{i,jmod} > l_1$  then
29                  $numExcessElectrons \leftarrow S_{i,jmod} - l_1$ ;
30               else
31                  $numExcessElectrons \leftarrow 0$ ;
32            $jmod \leftarrow j$ ;
33        $numExcessElectrons \leftarrow \frac{S_{ij} - l_1}{2}$ ;
34       while ( $numExcessElectrons > 0$ ) and ( $jmod \geq 0$ ) do
35          $S_{i,jmod} \leftarrow S_{i,jmod} - numExcessElectrons$ ;
36          $jmod \leftarrow jmod - 1$ ;
37         // Electrons reaching the edge of the CCD will not be detected
38         if  $jmod \geq 0$  then
39            $S_{i,jmod} \leftarrow S_{i,jmod} + numExcessElectrons$ ;
40           /* Make sure the pixel you move the excess electrons to does not get
41              saturated too */
42           if  $S_{i,jmod} > l_1$  then
43              $numExcessElectrons \leftarrow S_{i,jmod} - l_1$ ;
44           else
45              $numExcessElectrons \leftarrow 0$ ;

```

Table 2: Overview of which effects come into play for the serial pre-scan (bias register map) and the parallel over-scan (smearing map).

	Serial pre-scan (bias register map)	Parallel over-scan (smearing map)
Cosmics		✓
Photon noise		✓
Dark current		✓
Readout noise	✓	✓
Open-shutter smearing		✓
Digital saturation	✓	✓
Gain (CCD + FEE)	✓	✓
Electronic offset	✓	✓
Discretisation	✓	✓